

TOPAZ v2: A Continuous Topological Interpretation of Quantum Circuits with Parametric Trotter Expressions and Noise-Resilient Topology Optimization

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Abstract

The standard description of a quantum circuit is fundamentally discrete, imposing severe geometric constraints on continuous optimization landscapes. This paper introduces the Continuous Topological Interpretation (CTI), which conceptualizes circuits as continuous landscapes embedded in an extended parameter space, proves that quantum entanglement acts as a topological mechanism of manifold enlargement, and shows that noise can be understood as geometric deformation. To navigate these landscapes, we derive and validate the Parametric Trotter Expression (PTE) and its unitary projection (U-PTE), a weighted superposition of Hamiltonian exponentials that expands variational expressivity beyond fixed gate structures while restoring physical realizability via polar decomposition. Building on these foundations, we present TOPAZ v2, a noise-resilient topology optimization framework that pre-tunes ansatz parameters before execution. TOPAZ v2 introduces a dual-MMA optimiser decoupling intensity and timing subspaces, an $O(2^N)$ tensor-dot noise engine achieving a measured $\sim 3838\times$ speedup, and a hardware execution path validated on IBM’s 156-qubit Heron r2 processor. On a 12-qubit Jordan–Wigner LiH benchmark, TOPAZ achieves a mean energy error of 30.78 ± 0.97 mHa, beating Zero-Noise Extrapolation by $1.87\times$; on live hardware, Direct Fidelity Estimation shows a $10.71\times$ gain at $\sim 18\sigma$ in ~ 2 minutes of QPU time. Across every backend, the optimiser converges to the same hardware-transferable fidelity floor—direct empirical evidence for the “structural well” predicted by CTI.

Keywords: Continuous Topological Interpretation, Entanglement Geometry, Parametric Trotter Expression, Quantum Topology Optimization, Noise-Resilient Quantum Circuits, NISQ, IBM Heron, Structural Well, Method of Moving Asymptotes.

[Figure 1: Methodology Pipeline Flowchart — insert methodology_pipeline_v3.png]

Figure 1. The TOPAZ v2 research pipeline. Stage 1 initializes heterogeneous intensity (ρ) and timing (τ) parameters. Stage 2 computes the noiseless ideal state via the tensor-dot engine. Stage 3 runs pre-execution topology optimization (Dual-MMA over polar-projected U-PTE layers) and post-execution ZNE in parallel. The optimized parameters are finally validated on IBM Heron via Direct Fidelity Estimation.

1 Introduction

In classical computer science, the most fundamental abstraction is the instruction sequence: a list of discrete operations executed in a rigid, step-by-step order. Early quantum computing has largely inherited this discrete mental model. A quantum circuit is standardly defined as a finite sequence of unitary gates—such as single-qubit rotations and CNOTs—drawn from a discrete gate set and applied to a register of qubits over discrete time steps.

While this discrete, algebraic view is indispensable for building compilers and designing error-correcting codes, it creates significant bottlenecks when we attempt to optimize quantum circuits on physical hardware. In Variational Quantum Algorithms (VQAs) and Quantum Optimal Control (QOC), we optimize continuous physical parameters (such as laser pulse durations or gate rotation angles) to prepare a target quantum state or minimize the expectation value of an observable.

When restricted to the standard gate-based representation, these optimization pathways are forced to remain strictly within the curved, highly non-convex manifold of the unitary group $U(2^N)$. This constraint makes navigating the parameter space exceptionally difficult. The resulting optimization landscapes are often plagued by sharp energy cliffs, local minima, and “barren plateaus” where the gradient signals vanish exponentially with the size of the system, stalling convergence entirely.

This paper introduces a different conceptual and mathematical framework: the *Continuous Topological Interpretation (CTI)* of quantum circuits. Rather than treating a quantum circuit as a static, discrete sequence of rigid operations, we view it as a continuous, high-dimensional landscape. In this geometric framework, we embed the standard unitary group $U(d)$ as a constrained, physically realizable submanifold sitting inside a larger, unconstrained “ambient” space of general complex matrices, $GL(d, \mathbb{C})$.

The Continuous Geometric Framework	
Traditional Discrete	Proposed CTI Model
• Sequence of gates. • Restricted to $U(d)$. • Barren plateaus. • Correlation resource.	• Continuous landscape. • Explores $GL(d, \mathbb{C})$. • Unconstrained flow. • Manifold enlargement.

Figure 2. Comparison between traditional discrete gate-based models and the Continuous Topological Interpretation.

By allowing our optimization parameters to temporarily step off the rigid unitary path and explore this larger ambient space, the mathematical equations governing our gradients become remarkably smooth. Gradients can flow freely around localized energy barriers that would block a strictly unitary optimizer. Prior to physical implementation, we restore physical realizability by projecting the ambient operators back to the nearest unitary matrix using a mathematically clean polar projection.

By reframing quantum circuits as continuous land-

scapes, we establish three core theoretical results:

1. **Entanglement as Manifold Enlargement:** We prove that quantum entanglement has a clean geometric interpretation. It acts as a topological mechanism that physically enlarges the dimension of the accessible parameter manifold.
2. **Unitary Regularization:** We demonstrate that projecting ambient operators back onto the unitary group smooths the optimization landscape, stabilizing gradient signals.
3. **Noise Deformation:** We show that environmental noise can be viewed as a continuous vector field pulling the state landscape away from our target manifold.

Quantum error correction is not available on near-term hardware. What is available is a stack of quantum error mitigation (QEM) techniques — Zero-Noise Extrapolation (ZNE) [8], Probabilistic Error Cancellation, Clifford Data Regression, dynamical decoupling, and twirled read-out [7, 21] — all of which operate on circuit outputs after execution. They post-process measurements, fit error models, or extrapolate from amplified-noise runs.

TOPAZ v1 [2] introduced an additional layer that acts *before* the circuit runs. Given a parametric Trotter ansatz with intensity weights ρ and timing angles τ , the framework holds the circuit’s gate topology fixed and tunes (ρ, τ) so the noisy output is intrinsically closer to the ideal one. The v1 paper established the PTE parameterisation [3], the triple-channel correlated noise model, the specialised Method of Moving Asymptotes (MMA) adaptation [12], and 8-qubit validation under simulated noise. It also identified Grover’s algorithm as incompatible with the framework on topological grounds (continuity and convexity violations in the amplitude-amplification landscape).

This paper is the v2 follow-up. We retain the v1 mathematical structure and add what was needed to take the framework from simulator demonstrations to a calibrated device-noise model and to live hardware:

- A *dual-MMA* optimiser (Sec. 4.2.1) with two meanings: subspace separation between ρ and τ inside one module, and stacked MMA-over-PTE modules across parameter classes.
- A tensor-dot noise engine (Sec. 4.2.2) that brings Kraus-map evaluation from $O(2^{3N})$ to $O(2^N)$ per term, with explicit scope: where the engine runs and where it does not.
- A hardware execution path (Sec. 4.2.3). The SVD-of-sum unitary used in simulator runs does not transpile cleanly for $N \geq 6$; we replace it on hardware with a product-of-exponentials Trotter form that compiles to a shallow 21-gate circuit for the 8-qubit case.
- A 12-qubit LiH benchmark, a head-to-head against ZNE/CDR/VD on the same physical baseline (Sec. 5.3), a topological-lift result for Grover’s diffusion oracle (Sec. 5.4), and a real-hardware run on IBM’s 156-qubit Heron-r2 (Sec. 5.6).

- A direct empirical observation of the “structural well” predicted by the companion CTI paper [1]: the optimiser converges to the same fidelity floor from random initialisations, and the floor transfers across hardware (Sec. 6.2).

We retain the original positioning: TOPAZ is one layer of the QEM stack, not a stand-alone solution. Its job is to deliver a parameter setting that survives the device’s noise as well as possible before any post-execution mitigation runs. ZNE, DD, TREX and PEC consume the resulting circuit and benefit from its cleaner baseline.

Variational Quantum Algorithms (VQAs) represent the leading approach for near-term quantum computing but face fundamental expressivity limitations due to shallow circuit depths and restricted parameterization schemes. Traditional VQA circuits employ fixed gate structures with rotation parameters, constraining exploration to limited subspaces of the unitary group. In the NISQ era, where coherence times demand circuit compression, maximizing expressivity per gate operation becomes critical.

This work introduces and validates the Parametric Trotter Expression (PTE), a mathematical framework that expands variational circuit expressivity through weighted superpositions of quantum evolution primitives. Unlike conventional approaches maintaining strict unitarity, PTE enables exploration of expanded operator manifolds via temporary departure from the unitary group, followed by polar projection to the Unitary PTE (U-PTE) operator.

The contributions include: (1) theoretical derivation from Schrödinger evolution via Hamiltonian decomposition, (2) proof that weighted unitary superpositions require polar projection for physical realizability, (3) numerically stable U-PTE construction algorithms, and (4) quantum circuit objective functions.

This validation establishes theoretical foundations for PTE implementation in topology-optimized quantum architectures and provides rigorous basis for high objective quantum computing applications.

Note.. This paper is primarily designed to validate and verify the mathematical rigor of the Parametric Trotter Expression and subsequent operator constructions. Implementation studies, performance benchmarks, and practical applications are addressed in companion work.

2 Theory (CTI): A Continuous Topological Interpretation of Quantum Circuits and Entanglement

2.1 The Ambient Space and the Unitary Submanifold

To build the mathematical foundation of CTI, we must define the space in which our continuous quantum operators reside.

2.1.1 The Extended Operator Space

Let $\mathcal{H}_N \cong \mathbb{C}^d$ be the Hilbert space of an N -qubit system, where $d = 2^N$. The standard operations are represented by the unitary group:

$$U(d) = \{U \in \text{Mat}_d(\mathbb{C}) : U^\dagger U = I\} \quad (1)$$

This group is a compact, real Lie group of dimension d^2 . In CTI, we embed $U(d)$ as a constrained submanifold within $\text{GL}(d, \mathbb{C})$, or more generally, the full space of linear operators $\text{Mat}_d(\mathbb{C}) \cong \mathbb{C}^{d^2}$. We define a continuous circuit operator $A(\boldsymbol{\lambda})$ as a mapping:

$$A : \mathbb{R}^M \rightarrow \text{Mat}_d(\mathbb{C}) \quad (2)$$

where $\boldsymbol{\lambda}$ represents the continuous coordinates of our landscape. Because $A(\boldsymbol{\lambda})$ is not restricted to be unitary, operations can interpolate and scale.

2.1.2 The Geometric Projection Operator

To execute these operators, we must map them back to the physically realizable unitary submanifold via the **Polar Projection Operator**, $\mathcal{P}_U$.

Theorem 1 (Polar Projection). *Every invertible matrix $A \in \text{GL}(d, \mathbb{C})$ has a unique polar projection $\mathcal{P}_U(A)$ that maps it to the unique closest unitary matrix in the Frobenius norm:*

$$\mathcal{P}_U(A) = \arg \min_{U \in U(d)} \|A - U\|_F^2 \quad (3)$$

where $\|X\|_F = \sqrt{\text{Tr}(X^\dagger X)}$ is the Frobenius norm. *Proof.* By the polar decomposition theorem, any invertible A can be uniquely written as QP , where Q is unitary and $P = (A^\dagger A)^{1/2}$ is positive definite. Minimizing $\|A - U\|_F^2$ is equivalent to maximizing $\text{Re}(\text{Tr}(A^\dagger U))$. Substituting $A = QP$, we maximize $\text{Tr}(PQ^\dagger U)$. Let $Z = Q^\dagger U$. As P is positive definite with positive eigenvalues, $\text{Tr}(PZ)$ is maximized when $Z = I$, implying $U = Q$. Thus, $\mathcal{P}_U(A) = A(A^\dagger A)^{-1/2}$. ■

2.1.3 Unitary Regularization: Filtering the Landscape

This projection is the geometric bridge of the CTI framework. It guarantees that our algorithm can navigate smoothly in the unconstrained ambient space while ensuring the output $U_{\text{safe}}(\boldsymbol{\lambda}) = \mathcal{P}_U(A(\boldsymbol{\lambda}))$ is unitary. This process, **unitary regularization**, “pins” singular values back to 1.0, acting as a filter that smooths sharp wrinkles in the loss landscape and stabilizes gradient signals through deep circuits.

2.2 Quantum Entanglement as Manifold Enlargement

2.2.1 The “Paper Sheet” Metaphor

Let us visualize this using an intuitive geometric analogy. If we restrict our qubits to be unentangled, the states we can prepare form a thin, folded, low-dimensional sheet winding through the massive Hilbert space volume. We

The Entanglement Lift

$\mathcal{M}_{\text{unt}}$ (Restricted sheet) $\xrightarrow{\lambda_{\text{ent}}}$ $\mathcal{M}_{\text{ent}}$ (High-dim volume)

Adding entanglement is equivalent to inflating a sheet of paper into a broad volume. The extra spatial dimensions allow the gradient path to simply bend around energy barriers.

Figure 3. Visualizing entanglement as a dimensional inflation of the landscape.

call this the unentangled manifold, $\mathcal{M}_{\text{unt}}$. Navigating $\mathcal{M}_{\text{unt}}$ is restrictive. Adding entangling operations is equivalent to physically inflating this thin sheet into a high-dimensional volume: the entangled manifold, $\mathcal{M}_{\text{ent}}$. Suddenly, the optimizer has extra dimensions of freedom to bypass bottlenecks.

2.2.2 Formal Proof of Parameter-Space Enlargement

Suppose we have an N -qubit register partitioned into k disjoint clusters S_a . Without entanglement, operations are restricted to localized rotations within each cluster. For each cluster S_a , we define:

$$U_{S_{\text{local},a}}(\lambda_{\text{local},a}) = \bigotimes_{j \in S_a} \exp\left(-i \frac{\theta_{a,j}}{2} \sigma_{j,\alpha}\right) \quad (4)$$

The set of all preparable product states forms $\mathcal{M}_{\text{unt}}$:

$$\mathcal{M}_{\text{unt}} = \left\{ \bigotimes_{a=1}^k |\phi_a(\lambda_{\text{local},a})\rangle : \lambda_{\text{local},a} \in \mathbb{R}^{d_{\text{local},a}} \right\} \quad (5)$$

The dimension is bounded by $\dim(\mathcal{M}_{\text{unt}}) = \sum_{a=1}^k d_{\text{local},a}$. Now, introduce entangling parameters $\lambda_{\text{ent},a}$ that scale interaction terms bridging cluster boundaries. The combined circuit block dimension becomes $d_a = d_{\text{local},a} + d_{\text{ent},a}$. The dimension of the accessible landscape $\mathcal{M}_{\text{ent}}$ is now:

$$\dim(\mathcal{M}_{\text{ent}}) = \sum_{a=1}^k (d_{\text{local},a} + d_{\text{ent},a}) > \dim(\mathcal{M}_{\text{unt}}) \quad (6)$$

This confirms that **quantum entanglement physically inflates the dimension of our parameter space**, unlocking higher-dimensional navigation.

2.3 Optimizing over the Ambient Landscape

In traditional algorithms, optimization is performed directly on $U(d)$. This requires calculating gradients along curved directions. In CTI, we open up the ambient space $GL(d, \mathbb{C})$, where operations are linear and gradients are smooth. This unconstrained exploration allows CTI to navigate complex landscapes with speed. The polar projection acts as a guide rail, ensuring valid physical states while the ambient space provides the freedom to find the shortest pathway.

The Unconstrained Gradient Path

$GL(d, \mathbb{C})$ (Smooth landscape) $\xrightarrow{\nabla}$ Ambient Step

$\xrightarrow{\mathcal{P}_U}$ Physical Unitary on $U(d)$

By allowing the optimizer to temporarily fly through non-unitary space, we bypass coordinate bottlenecks of the curved manifold.

Figure 4. The optimization pathway in CTI leverages non-unitary space.

2.4 Noise as Geometric Deformation

We visualize noise as a physical deformation of our state landscape. Environmental noise (T_1, T_2) acts as an unwanted **continuous vector field** pulling the state toward high-entropy interiors. Instead of adding qubits,

The Noise Deformation Landscape

Ideal Well $\xrightarrow{\text{Noise field}}$ Deformed Well

$\xrightarrow{\text{Topology Opt}}$ Structural Well

We use continuous parameters to shape a “structural shield” that cancels out the noise vector field.

Figure 5. Visualizing noise as deformation and mitigation as landscape shaping.

we use coordinates λ to search for a **structural well** where the noise field and control gradients cancel. By pre-tuning parameters to sit inside this well, we build resilience directly into the circuit’s structural parameters.

3 The Operator (PTE/UPTE): Derivation and Validation of the Parametric Trotter Expression

3.1 Background

3.1.1 Motivation and Theoretical Foundation

The Parametric Trotter Expression (PTE) is a parameterized operator ansatz where the free variables correspond to gate intensities and timings. The key purpose of PTE is not to automatically enforce unitarity, but to provide a flexible ansatz that allows an optimizer to explore the operator-level control landscape with unprecedented expressivity.

Theoretical Insight: PTE uses weighted superpositions of candidate sub-circuits, significantly expanding the reachable manifold beyond traditional variational quantum circuits. However, this expressivity comes at a cost: The resulting operators may stray outside the unitary manifold. The projection to the nearest unitary (the U-PTE operator) acts as a mathematical safeguard, not as the original motivation for introducing PTE.

The theoretical foundation rests on the observation that quantum hardware trends toward larger qubit counts but shallower coherence windows. PTE addresses circuit depth by allowing weighted, timing-aware superpositions that approximate target unitaries with fewer gate layers.

3.1.2 From Schrödinger Equation to Time Evolution

Mathematical Foundation: The time-dependent Schrödinger equation governs quantum dynamics:

$$-i\hbar \frac{\partial}{\partial t} |\Psi(t)\rangle = H |\Psi(t)\rangle \quad (7)$$

Formal Solution: Integrating this differential equation yields:

$$|\Psi(t)\rangle = U(t) |\Psi(0)\rangle, \quad U(t) = e^{-iHt/\hbar} \quad (8)$$

Setting $\hbar = 1$ for convenience:

$$U(t) = e^{-iHt} \quad (9)$$

Theoretical Significance: The exponential operator e^{-iHt} encodes continuous time evolution. Simulating quantum dynamics means approximating this operator, establishing the mathematical bridge between continuous evolution and discrete approximation schemes.

3.1.3 Hamiltonian Decomposition and Non-Commutativity

Mathematical Structure: We decompose the Hamiltonian into Hermitian terms:

$$H = \sum_{j=1}^M H_j, \quad H_j = H_j^\dagger \quad (10)$$

Ideal Case: If all terms commute, $[H_j, H_k] = 0$ for all j, k , then:

$$e^{-iHt} = \prod_{j=1}^M e^{-iH_j t} \quad (11)$$

Reality: In practice, Pauli operators generally do not commute. For example:

$$[\sigma_x, \sigma_y] = 2i\sigma_z \neq 0 \quad (12)$$

Theoretical Consequence: This non-commutativity reflects fundamental quantum incompatibility. The commutator $2i\sigma_z$ represents the inherent quantum mechanical nature that prevents direct factorization, necessitating approximation techniques.

3.1.4 Trotter-Suzuki Approximation Theory

Mathematical Construction: Split time t into n slices with $\delta = t/n$:

$$A = \exp(-iHt) = \lim_{n \rightarrow \infty} \left[\prod_{p=1}^n \prod_j \exp(-iH_j \delta_j) \right] \quad (13)$$

NOTE: At finite n , it is an approximation, with greater n showing closer approximations

Error Analysis: For finite n , the approximation error is:

$$e^{-iHt} - A = \exp(-iHt) - \lim_{n \rightarrow \infty} \left[\prod_{p=1}^n \prod_j \exp(-iH_j \delta_j) \right] \quad (14)$$

NOTE: Again, at finite n , it is an approximation, with greater n showing closer approximations

Theoretical Foundation: The Trotter formula provides a systematic way to handle non-commuting exponentials by discretizing time evolution. The error term $O(t^2/n)$ quantifies the trade-off between accuracy and circuit feasibility—as n grows, accuracy improves but circuit depth increases.

3.2 Methods

3.2.1 Introducing Design Variables: The Paradigm Shift

Innovation: I extend the traditional approximation by introducing two classes of design variables, inspired by variational quantum algorithms:

- **Intensity weights:** $\rho_j \in [0, 1]$
- **Timing scaling factors:** $\tau_j \in [0, \tau_{\max}]$

Notation Clarification: We distinguish between the base Trotter time slice $\delta = t/n$ and the dimensionless timing design variables $\tau_{j,k} \in [0, \tau_{\max}]$. The actual evolution time for each term is $\delta \cdot \tau_{j,k}$, allowing the optimizer to scale individual gate durations within physical bounds determined by coherence constraints.

Two Theoretical Approaches:

(A) Hamiltonian-weighted exponential (always unitary):

$$H_\rho = \sum_j \rho_j H_j, \quad U = e^{-iH_\rho \delta} \quad (15)$$

(B) Sum of exponentials (PTE ansatz):

$$B_k(\rho, \tau) = \sum_j \rho_{j,k} e^{-iH_j \delta \tau_{j,k}} \quad (16)$$

$$A(\rho, \tau) = \prod_{k=1}^n B_k(\rho, \tau) \quad (17)$$

The Parametric Trotter Expression: Approach (B) defines our central object:

$$A(\rho, \tau) = \prod_{k=1}^n \left[\sum_{j=1}^M \rho_{j,k} e^{-iH_j \delta \tau_{j,k}} \right] \quad (18)$$

Theoretical Distinction: Route (A) maintains unitarity by construction but limits expressivity to weighted Hamiltonians. Route (B) creates the PTE as a product of weighted sums, offering unprecedented expressivity for optimization at the cost of potential non-unitarity.

3.2.2 Mathematical Architecture of the Parametric Trotter Expression

Design Philosophy: Each exponential $e^{-iH_j \delta \tau_{j,k}}$ represents a candidate primitive unitary operation. The weights $\rho_{j,k}$ allow the optimizer to interpolate among

these primitives, creating a continuous space of approximately unitary operations.

Theoretical Power: This formulation decouples intensity and timing variables, enabling a single template to emulate many distinct quantum evolutions. The optimizer can discover optimal combinations that would be impossible to find in traditional parameter spaces constrained to remain unitary.

Mathematical Consequence: The sum structure within each layer breaks unitarity but enables the exploration of a vastly expanded operator manifold, requiring subsequent projection back to the unitary group.

3.2.3 Proof: Non-Unitarity of Weighted Unitary Sums

Theorem 1. A weighted sum of unitary operators is generally non-unitary.

Proof: Let $B = \sum_j \alpha_j U_j$ where U_j are unitary and α_j are complex weights. Then:

$$B^\dagger B = \sum_{j,k} \alpha_j^* \alpha_k U_j^\dagger U_k \quad (19)$$

For unitarity, we require $B^\dagger B = I$. This condition becomes:

$$\sum_{j,k} \alpha_j^* \alpha_k U_j^\dagger U_k = I \quad (20)$$

Analysis: This equation is satisfied only under highly restrictive conditions:

- All but one $\alpha_j = 0$ (trivial case)
- The U_j form an orthogonal set with specific weight relationships
- Special symmetric constructions with commuting unitaries

Counterexample: Consider two non-commuting unitaries U_1, U_2 with $\alpha_1 = \alpha_2 = \frac{1}{2}$. Then:

$$B^\dagger B = \frac{1}{4}(U_1^\dagger U_1 + U_1^\dagger U_2 + U_2^\dagger U_1 + U_2^\dagger U_2) = \frac{1}{4}(2I + U_1^\dagger U_2 + U_2^\dagger U_1) \neq I \quad (21)$$

Theoretical Implication: The PTE's expressive power directly stems from this non-unitarity. The mathematical freedom to leave the unitary manifold enables the exploration of operator spaces unreachable by traditional variational circuits.

3.2.4 Polar Decomposition: The Mathematical Bridge

Theorem 2. Every complex matrix A has a unique polar decomposition:

$$A = QP \quad (22)$$

where Q is unitary and P is positive semidefinite.

Explicit Construction:

$$P = (A^\dagger A)^{1/2}, \quad Q = A(A^\dagger A)^{-1/2} \quad (23)$$

SVD Implementation: Using singular value decomposition $A = W\Sigma V^\dagger$:

$$Q = WV^\dagger \quad (24)$$

Mathematical Property: Q is the closest unitary matrix to A in the Frobenius norm:

$$Q = \arg \min_{U \text{ unitary}} \|A - U\|_F \quad (25)$$

Theoretical Significance: Polar decomposition provides the mathematical mechanism to project any matrix back to the unitary manifold while preserving maximum similarity to the original operator. For quantum applications, efficient algorithms leverage circuit structure for improved numerical stability.

3.2.5 Constructing the U-PTE Operator

Layer-wise Projection Algorithm: For each layer k :

- Compute the weighted sum: $B_k = \sum_{j=1}^M \rho_{j,k} e^{-iH_j \delta \tau_{j,k}}$
- Apply polar projection: $U_k = B_k (B_k^\dagger B_k)^{-1/2}$

Final Assembly:

$$U_{\text{safe}}(\rho, \tau) = \prod_{k=1}^n U_k \quad (26)$$

Mathematical Guarantee: Each U_k is strictly unitary, and products of unitary operators remain unitary. Thus, U_{safe} is guaranteed to be a valid quantum operator.

Theoretical Advantage: Layer-wise projection maintains numerical stability better than attempting to project the full product $A(\rho, \tau)$ after construction. Each intermediate step remains well-conditioned, preventing the accumulation of numerical errors.

3.3 Results

3.3.1 Operator Fidelity: Measuring Quantum Similarity

Mathematical Definition: The operator fidelity measures closeness between target and approximate unitaries using the normalized Hilbert-Schmidt inner product:

$$\langle U_{\text{target}}, U_{\text{approx}} \rangle = \frac{1}{d} \text{Tr}(U_{\text{target}}^\dagger U_{\text{approx}}) \quad (27)$$

where d is the Hilbert space dimension.

Fidelity Measure:

$$F_{\text{op}} = |\langle U_{\text{target}}, U_{\text{approx}} \rangle|^2 = \frac{1}{d^2} |\text{Tr}(U_{\text{target}}^\dagger U_{\text{approx}})|^2 \quad (28)$$

Theoretical Interpretation: This quantity represents the probability that the approximate operation agrees with the target over arbitrary quantum states—a fundamental measure of operational equivalence in quantum mechanics.

3.3.2 State Dependence and Parameter Sensitivity

State Evolution: Given a reference state $|\psi_0\rangle$, the prepared state is:

$$|\psi(\rho, \tau)\rangle = U_{\text{safe}}(\rho, \tau)|\psi_0\rangle \quad (29)$$

State Fidelity: For a target state $|\psi_{\text{target}}\rangle$:

$$F_{\text{st}} = |\langle\psi_{\text{target}}|\psi(\rho, \tau)\rangle|^2 = |\langle\psi_{\text{target}}|U_{\text{safe}}(\rho, \tau)|\psi_0\rangle|^2 \quad (30)$$

Gradient Analysis: For small parameter changes:

$$\Delta|\psi\rangle \approx (\partial_x U_{\text{safe}})|\psi_0\rangle \Delta x \quad (31)$$

Mathematical Challenge: Computing $\partial_x U_{\text{safe}}$ requires careful differentiation through the polar decomposition step, involving derivatives of matrix exponentials and the inverse square root operation through the product structure.

3.3.3 Numerical Stability and Implementation

Critical Consideration: Computing $(B_k^\dagger B_k)^{-1/2}$ can become numerically unstable if $B_k^\dagger B_k$ is ill-conditioned.

Stable Methods:

- Direct SVD: Most robust for small systems
- Newton-Schulz iteration: Iterative method with quadratic convergence
- Quantum polar decomposition algorithms: Specialized methods for quantum applications

Verification Protocol: Always check unitarity numerically:

$$\|U_k^\dagger U_k - I\|_F < \epsilon \quad (32)$$

where ϵ is a problem-appropriate tolerance.

3.3.4 SU(d) Normalization and Physical Equivalence

Mathematical Adjustment: To ensure membership in SU(d):

$$U_{\text{safe}} \rightarrow U_{\text{safe}} \cdot (\det U_{\text{safe}})^{-1/d} \quad (33)$$

Physical Significance: This normalization removes global phases that are mathematically present but physically irrelevant. The determinant condition $\det U = 1$ ensures consistency with the special unitary group without affecting quantum mechanical predictions.

3.3.5 Optimization Framework and Objective Functions

Primary Objectives:

Operator Fidelity Loss:

$$L_{\text{op}}(\rho, \tau) = 1 - \frac{1}{d^2} |\text{Tr}(U_{\text{target}}^\dagger U_{\text{safe}}(\rho, \tau))|^2 \quad (34)$$

State Fidelity Loss:

$$L_{\text{st}}(\rho, \tau) = 1 - |\langle\psi_{\text{target}}|U_{\text{safe}}(\rho, \tau)|\psi_0\rangle|^2 \quad (35)$$

Regularized Loss: To encourage near-unitarity during optimization:

$$L(\rho, \tau) = L_{\text{primary}}(\rho, \tau) + \lambda \|A(\rho, \tau)^\dagger A(\rho, \tau) - I\|_F^2 \quad (36)$$

Theoretical Balance: The regularization parameter λ controls the trade-off between expressivity (allowing exploration of non-unitary space) and physical validity (staying close to unitary operators).

Optimization Methods: The PTE framework can be embedded within various optimization algorithms. The Method of Moving Asymptotes (MMA) provides a particularly suitable approach due to its convex approximation strategy and proven effectiveness in structural optimization problems.

4 The Framework (TOPAZ v2): Noise-Resilient Topology Optimization

4.1 Background: the PTE ansatz

We summarise only what is needed for the rest of the paper; the full derivation, the proof that the weighted sum is generically non-unitary, the polar projection construction, and the numerical-stability analysis are in the PTE companion paper [3].

For an N -qubit register and a Hamiltonian $H = \sum_{j=1}^M H_j$ decomposed into Hermitian (typically two-local Pauli) terms, the layer- k PTE operator is

$$B_k(\rho, \tau) = \sum_{j=1}^M \rho_{j,k} e^{-iH_j \delta \tau_{j,k}}, \quad (37)$$

with intensities $\rho_{j,k} \in [0, 1]$ (normalised to $\sum_j \rho_{j,k} = 1$), timings $\tau_{j,k} \in [0, \tau_{\text{max}}]$, and a fixed base time slice δ . Equation (37) is generically non-unitary. The layer-wise polar projection

$$U_k = B_k (B_k^\dagger B_k)^{-1/2}, \quad U_{\text{safe}}(\rho, \tau) = \prod_{k=1}^L U_k \quad (38)$$

returns the unique nearest unitary to B_k in Frobenius norm. We compute it by SVD with a Newton-Schulz refinement step and a final $U_{\text{safe}} \rightarrow U_{\text{safe}} \cdot (\det U_{\text{safe}})^{-1/d}$ to remove the global phase. The Trotter error of the underlying decomposition scales as $O(t^2/n)$ [16, 22].

The reason for using PTE at all, rather than a standard rotation-angle parameterisation, is expressivity: the weighted superposition of exponentials reaches operators that a fixed-structure variational ansatz cannot. The polar projection then snaps the result back to the unitary manifold so the circuit can be physically realised. See [3] for the proof that this expansion is strict.

4.2 The TOPAZ v2 framework

4.2.1 Dual-MMA optimiser

The MMA algorithm [12, 13] constructs, at each iteration, a strictly convex separable approximation of the objective and solves it analytically subject to asymptotic bounds

$[L_i, U_i]$. v1 used a single MMA instance over the concatenated (ρ, τ) vector. v2 introduces a two-meaning “dual” that we want to spell out, because the two meanings travel together but are not the same thing.

Meaning 1: subspace separation within one MMA module. The intensity vector ρ and the timing vector τ are heterogeneous parameter classes whose finite-difference step sizes alone differ by roughly four orders of magnitude ($\epsilon_\rho=10^{-4}$ vs. $\epsilon_\tau=\pi/4$), and whose gradient magnitudes during optimisation differ by similar factors. A single MMA instance with a shared move-limit budget and shared asymptote schedule systematically misallocates the parameter-update budget. Inside any single MMA module, we therefore run two independent MMA solvers — one for ρ , one for τ — with their own asymptote, move-limit, and momentum state.

Meaning 2: stacked MMA modules across parameter classes. The structural layer described above tunes (ρ, τ) for noise resilience. Many target algorithms carry an additional class of parameters that would normally be tuned by a classical optimiser (COBYLA, SPSA, Adam) — the rotation angles of a VQE ansatz, for example. TOPAZ v2 replaces every such classical optimiser with a second MMA module, wrapped around its own small PTE/U-PTE ansatz built over *only those parameters*, not over (ρ, τ) . The two modules do not share gradient state, asymptote schedules, or move limits; they alternate, each treating the other module’s parameters as fixed during its own update step.

The principle generalises: any heterogeneous parameter class that would otherwise live inside a single classical optimiser can be promoted to its own MMA-over-PTE module. The current paper exercises two modules (structural + algorithmic), which is where the name “dual” comes from. A third class (say, measurement-basis angles for an adaptive scheme) would slot in the same way.

Why MMA, and not Adam / SPSA / COBYLA, for the second module. The choice of MMA for the algorithmic module is not interchangeable with off-the-shelf variational optimisers. The two modules alternate updates against the same noisy objective; for the architecture to converge, each module’s update has to be *compatible* with the surrogate the other module is implicitly carrying. MMA’s separable convex-approximation step, with its asymptotic bounds and conservative move limits, produces updates that lie inside a region the alternating module has not yet ruled out.

We have the negative-result ablation to back this up. Replacing the algorithmic module with a vanilla VQE solver (Adam-class or SPSA-class) while keeping the structural TOPAZ/U-PTE+MMA layer in place produced energies off by $\gtrsim 5$ Ha from FCI on LiH — not catastrophic divergence (no NaNs, the loop terminates), but useless by chemistry standards and roughly two orders of magnitude worse than the ~ 30 mHa achieved with two MMA modules. The structural and algorithmic layers individually behave reasonably under their native optimisers; what fails is the *composition*. Carrying MMA

through to the second module is what makes the two layers’ updates complement rather than fight each other. We do not claim a formal proof of this — it is an empirical architectural constraint — but the gap between the two configurations is large enough that the constraint is not subtle.

Where each meaning applies in this paper. Meaning 2 (Stacked Modules) is utilized for the VQE-based LiH benchmark to navigate the interplay between structural and algorithmic parameters. For the 8-qubit hardware validation, which focuses on a fixed-Hamiltonian Trotter evolution, we utilize only Meaning 1 (Subspace Separation). This isolates the effectiveness of TOPAZ in establishing a resilient physical baseline without the added noise variance of an alternating algorithmic search.

Mechanics inside each module. Each MMA solver, in either module, retains the v1 mechanics. Gradients use finite differences with $\epsilon_\rho=10^{-4}$ and $\epsilon_\tau=\pi/4$ (or the equivalent step sizes for the algorithm-layer parameters), smoothed with momentum coefficient $\alpha = 0.4$:

$$\nabla_{\text{smooth}} = \alpha \nabla_{\text{cur}} + (1 - \alpha) \nabla_{\text{prev}}. \quad (39)$$

Move limits adapt online: $\Delta > 10^{-2}$ expands to 0.6 (aggressive search); $10^{-3} < \Delta < 10^{-2}$ stays neutral; $\Delta < 10^{-3}$ contracts to 0.15 (refinement). Asymptotes expand by $1.2\gamma(x_i - L_i)$ in productive iterations and contract by $\gamma(x_i - L_i)$ during stagnation, with $\gamma = 0.5$.

The objective is the noisy fidelity (or, for chemistry, the noisy energy expectation) of $U_{\text{safe}}(\rho, \tau) |\psi_0\rangle$ against an ideal target. A regularisation term

$$L(\rho, \tau) = L_{\text{prim}}(\rho, \tau) + \lambda \sum_{k=1}^L \left\| B_k^\dagger B_k - I \right\|_F^2, \quad \lambda = 10^{-3} \quad (40)$$

keeps the pre-projection operator close to unitarity and stabilises the polar step. In the two-module setup, each module carries its own regularisation term over its own PTE ansatz.

4.2.2 Tensor-dot engine and Mathematical Derivation

The tensor-dot engine was built to speed up one specific operation: applying a sequence of small (typically two-local) gates or Kraus operators to a dense state or density-matrix tensor without materialising the full $2^N \times 2^N$ Kronecker product. It is a wall-clock optimisation, not a complexity-theoretic separation.

The core efficiency of the engine relies on reframing the superoperator action $\rho \rightarrow \sum_a K_a \rho K_a^\dagger$ as a series of local tensor contractions. In a dense simulation, construction of the full superoperator scales as $O(2^{3N})$ because it treats the density matrix as a single block. We instead treat ρ as a tensor with $2N$ indices, where each index pair (i_n, j_n) corresponds to the state of the n -th qubit. For a k -local Kraus operator $K^{(a)}$ acting on a subset of qubits $\mathcal{S}$ (e.g., a two-qubit gate), the update rule is:

Table 1. Per-evaluation runtime at $N = 10$ for one layered triple-channel Kraus map. Same workstation, same Kraus operators, no GPU. The wall-clock speedup is what makes 10-seed Monte-Carlo sweeps tractable on the legacy triple-channel noise path (Sec. 5.2 and Sec. 5.4). On the **FakeFez** simulator path, the engine is used only for the noiseless target state; on real **ibm_fez**, it is bypassed entirely (see scope discussion below).

Engine	Time (s)	Complexity per term
Qiskit dense Kraus map	180.400	$O(2^{3N})$
Tensor-dot (this work)	0.047	$O(2^N)$
Speedup	$\sim 3838\times$	—

$$\rho'_{i'_S, i_{Sc}; j'_S, j_{Sc}} = \sum_a K_{i'_S, i_S}^{(a)} \rho_{i_S, i_{Sc}; j_S, j_{Sc}} K_{j_S, j'_S}^{(a)\dagger} \quad (41)$$

where S^c denotes the set of spectator qubits. By performing the contraction locally, we avoid constructing the $2^{2N} \times 2^{2N}$ superoperator. The computational complexity for a k -local gate becomes $O(2^N)$. practically, for a 10-qubit system, the engine performs roughly 2^{10} operations per term instead of 2^{30} , allowing for the massive wall-clock acceleration required for multi-seed optimization. Per-evaluation runtime is given in Table 1.

Scope: where it is used and where it is not. The engine plays different roles on different experiment paths, and we want to be explicit about it because it has been a source of confusion:

- **Legacy triple-channel path (LiH, algorithm tests).** The engine handles both the ideal-state computation and the noisy density-matrix evolution. This is the case Table 1 benchmarks.
- **FakeFez simulator path (Secs. 5.5–6.1).** The engine’s role is narrowed to the noiseless target state $|\psi_{\text{ideal}}\rangle$ only. It applies each 4×4 gate via tensor contraction on the rank-8 state tensor rather than building the full 256×256 product. All noisy density-matrix evaluations on this path are delegated to Qiskit’s **AerSimulator** density-matrix mode with the **FakeFez** noise model loaded; the engine has no role in the noisy branch.
- **Real **ibm_fez** hardware (Sec. 5.6).** The engine has *no* role on the live device. The physical backend executes the transpiled gates directly, and fidelity is measured by Direct Fidelity Estimation (DFE) on the QPU.
- **DFE evaluation (both simulator and hardware).** DFE uses a single function (`dfe_fidelity`) that runs a physical Qiskit circuit comprising the ansatz and the inverse target. This is independent of the classical tensor-dot engine on both paths.

4.2.3 Hardware execution

The simulator path of v1 builds U_{safe} as one $2^N \times 2^N$ matrix and applies it to the density matrix. That route is fine in simulation but does not survive transpilation: compiling a dense N -qubit unitary into a hardware-native

gate set is computationally intractable for $N \geq 6$ and produces circuits far deeper than necessary.

For hardware runs we substitute a first-order Trotter product,

$$U_{\text{circ}}(\rho, \tau) = \prod_{j=1}^M e^{-i \rho_{n,j} H_j \delta \tau_j}, \quad (42)$$

in which each factor is a 4×4 unitary on two adjacent qubits and is submitted as a separate two-qubit gate. For an 8-qubit ansatz with $M=21$ nearest-neighbour Pauli terms (seven $\{XX, YY, ZZ\}$ triples), Eq. (42) compiles to a shallow 21-gate circuit.

The product form in Eq. (42) is a first-order Trotter approximation of the ideal operator; it is not algebraically identical to the polar-projected unitary in Eq. (38). We acknowledge this as an “equivalence gap”: the optimizer finds a structural well in the landscape of U_{safe} , but we deploy it on the landscape of U_{circ} . We hypothesize that the noise-resilient basin is a fundamental feature of the underlying Hamiltonian evolution that remains invariant under first-order Trotterization, provided the Trotter error remains below the device’s noise floor. A rigorous characterization of how the structural well deforms between these two manifolds is a priority for future theoretical work.

The noisy density matrix on this path is computed entirely by **AerSimulator**: the evaluator builds the circuit of Eq. (42), appends a `save_density_matrix` instruction, transpiles, and submits the circuit with a calibrated device-noise model loaded (**FakeFez** for the experiments in Sec. 5, **FakeMontrealV2** for the earlier port). We handle Qiskit’s little-endian qubit convention with a single bit-reversal permutation at every density-matrix exchange.

4.2.4 Optimization Workflow and Algorithmic Implementation

The framework follows a hybrid digital-twin optimization path illustrated in the methodology pipeline (Fig. 1). We first establish a classical ground-truth using the $O(2^N)$ Tensor-Dot engine (Stage 2). We then run two mitigation tracks in parallel: a pre-execution track (TOPAZ) that finds noise-resilient coordinates in the parameter manifold via Dual-MMA and Polar Projection, and a post-execution track (ZNE). These are synthesized into a layered stack and finally validated through Direct Fidelity Estimation (DFE) on the live 156-qubit Heron processor. The core algorithmic implementation is detailed in Procedure 1.

```

PROCEDURE TOPAZ_v2_Research_Pipeline(H_target, problem_type, gate_topology, N_qubits, L_layers,
    M_terms):
    # STAGE 1: Initialization (SETUP)
    rho = random_matrix(L_layers, M_terms, range=[0,1], normalized_over_j=True)
    tau = random_matrix(L_layers, M_terms, range=[0, tau_max])
    iter = 0; max_iter = 40; stagnation_counter = 0; prev_fidelity = 0.0
    asymptotes_rho, move_limits_rho = initialize_MMA(); asymptotes_tau, move_limits_tau =
        initialize_MMA()

    # STAGE 2: Reference State computation
    psi_ideal = compute_ideal_state(H_target, gate_topology)

    # STAGE 3: Optimization Loop (while iter < 40 and delta_fid > threshold and stag < 5)
    while (iter < max_iter) and (stagnation_counter < 5):
        # 3.1 Build UPTE operator and apply layer-wise Polar Projection
        U_safe = identity_matrix(2^N_qubits)
        for k = 1 to L_layers:
            # delta (base time slice) is fixed and not optimized
            B_k = sum(rho[k][j] * matrix_exp(-i * H_j * delta * tau[k][j]) for j in M_terms)
            reg_k = lambda * frobenius_norm( (B_k.H @ B_k) - I )^2
            U_k = W @ V.H where (W, Sigma, V) = SVD(B_k) # Snap to nearest unitary
            U_safe = U_k @ U_safe

        # 3.2 Evaluation Track
        noisy_dm = AerSimulator.run(U_safe, noise_model="FakeFez")
        if problem_type == "LiH":
            loss_primary = abs(trace(noisy_dm @ H_target) - E_FCI)
        else:
            loss_primary = 1 - Fidelity(noisy_dm, psi_ideal)
        total_loss = loss_primary + sum(reg_k) # Penalizes non-unitary pre-projection operators

        # 3.3 Gradient and Update (Dual-MMA Meaning 1)
        # Finite-diff eps_rho = 1e-4, eps_tau = pi/4
        grad_rho = finite_diff(rho, total_loss); grad_tau = finite_diff(tau, total_loss)
        grad_smooth = smooth_gradients(grad, alpha=0.4)
        rho, tau = MMA_update(rho, tau, grad_smooth)

        # 3.4 Convergence tracking (delta_fid = abs(F_current - F_prev))
        if fidelity_change < 1e-3: stagnation_counter += 1
        else: stagnation_counter = 0; prev_fidelity = F_current; iter += 1
    # Convergence reached: Optimal parameters submitted to ibm_fez for DFE validation.

```

Listing 1. Pseudocode for the TOPAZ v2 Pipeline implementation. Note that δ is a fixed base time slice and is not optimized. Fidelity is defined as $F = |\langle \psi_{\text{ideal}} | \psi_{\text{noisy}} \rangle|^2$ for pure states and $F = \langle \psi_{\text{ideal}} | \rho_{\text{noisy}} | \psi_{\text{ideal}} \rangle$ for density matrices. For LiH chemistry, the primary loss is $|\langle H \rangle_{\text{noisy}} - E_{\text{FCI}}|$ instead of $1 - F$.

4.3 Noise models

4.3.1 Triple-channel correlated noise (legacy)

For the 8-qubit trajectory experiments of Sec. 5.1 and the LiH benchmark of Sec. 5.2 we use the v1 noise model: three sequential channels (depolarising at $p = 0.005$, amplitude damping at $\gamma = 0.008$, phase damping at $\lambda = 0.012$), each with both single-qubit and multi-qubit components. Multi-qubit correlations decay exponentially with qubit distance,

$$p_{\text{corr}}(d) = \alpha p e^{-d/\xi}, \quad \xi = 2.0, \quad (43)$$

with $\alpha \in \{0.30, 0.25, 0.25\}$ for depolarising, amplitude, and phase channels. The Kraus operators are standard [23, 24]. This model captures the qualitative shape of NISQ noise but is not tied to any specific device.

4.3.2 FakeFez

For the headline experiments (Secs. 5.5–5.6) we use IBM’s **FakeFez**, a calibrated noise snapshot of the live **ibm_fez** processor (Heron r2, 156 qubits). **FakeFez** carries per-gate error rates, qubit-specific T_1 and T_2 times, readout errors, and the device coupling map [15]. All experiments in Sec. 5.5 use **FakeFez** as the noise model during optimisation; the live-hardware run in Sec. 5.6 runs the optimised parameters on **ibm_fez** itself.

4.4 Algorithmic compatibility

Whether continuous parameter optimisation can work on a given quantum algorithm depends on four topological properties of its fidelity landscape: connectedness (gradient information has to flow), compactness (the search space has to be bounded), continuity (small parameter changes produce small fidelity changes), and

convexity (at least locally, gradients have to point toward improvements). Algorithms satisfying all four are natural TOPAZ targets; those violating one or more either need to be lifted (Sec. 5.4) or are not appropriate targets. Concretely: Grover as a whole algorithm is incompatible (violates continuity and convexity); the Grover *oracle layer* is compatible via the topological lift ($3.09\times$ gain, Sec. 5.4); Bernstein–Vazirani is compatible ($1.85\times$); VQE, VQA, QAOA, and parametrised quantum simulation satisfy all four properties by construction and benefit straightforwardly; quantum annealing and phase estimation are not appropriate targets (no parametrised classical control, or only fixed-structure parameters).

Variational algorithms [7] satisfy all four properties by construction. TOPAZ enhances them by acting at a lower level: before the classical optimiser starts, TOPAZ pre-tunes the ansatz for noise resilience, giving the downstream optimiser cleaner gradients and faster convergence.

4.5 Limitations

- **Classical simulation wall.** The optimisation loop relies on dense density-matrix simulation for gradient evaluation. For $N \gtrsim 20$, the 4^N memory footprint is infeasible. The planned path forward is a matrix-product-state extension of the tensor-dot engine; we have not implemented it.
- **Path mismatch on hardware.** As discussed in Sec. 5.6, optimisation against FakeFez’s reported quietest path can disagree with the device’s run-time path selection. Re-optimising on the live path is the immediate next experiment.
- **LiH not yet run under FakeFez.** The 10/12-qubit LiH results use the legacy triple-channel noise model. Re-running under FakeFez or `ibm_fez` conditions is planned.
- **Hessian Curvature.** The existence of the structural well is currently demonstrated empirically via trajectory convergence. A rigorous Hessian analysis to quantify curvature and verify local stability is planned.
- **On-Hardware Gradients.** Scaling structural optimization to $N > 20$ will require matrix-product-state extensions or hardware-assisted evaluation via the parameter-shift rule.

5 Results

5.1 8-qubit trajectories under triple-channel noise

These are the v1 simulator results, kept for continuity. Configuration: 8 qubits, 9 nearest-neighbour Pauli terms, 18 parameters, 40-iteration cap, stagnation-counter convergence. Pre-optimisation noisy fidelity ranges 0.07–0.70 depending on the random circuit instance, against a theoretical ideal of 0.95–0.98 (a 35–93 percentage-point degradation). Post-MMA fidelity rises to 0.30–0.88, an absolute improvement of 18–23 pp and a relative improvement of 24–80%.

[Figure: insert fakefez_trajectory_v3.png]

Figure 6. Two representative 8-qubit optimisation trajectories from independent random seeds, under triple-channel correlated noise. Both runs use the same code and the same noise model; the difference is the random (ρ, τ) initialisation. The plateau at ~ 0.37 (Run 1) and ~ 0.84 (Run 2) corresponds to two different local minima of the same landscape.

Figure 6 shows two representative trajectories from independent seeds. Run 1 (conservative): initial fidelity 0.172, final 0.371 (+19.86 pp), convergence at iteration ~ 26 , first-iteration gain +17.13 pp. Run 2 (aggressive): initial 0.233, final 0.838 (+60.41 pp), convergence at iteration ~ 21 , first-iteration gain +45.97 pp. Both runs are monotonic after the plateau onset; no oscillation is observed. The ~ 47 pp spread in final fidelity is what the dual-MMA optimiser is navigating: a multi-modal landscape with several local optima at different floor depths. In practice we run multiple seeds and keep the best, which is standard in variational quantum algorithm work [7, 25].

5.2 12-qubit LiH ground state

We push past the 4-qubit active-space tests of v1 to the full Jordan–Wigner mapping at 12 qubits, bond distance 1.6 Å, under the triple-channel correlated noise of Sec. 4.3. This is the experiment that exercises both meanings of the dual-MMA architecture (Sec. 4.2.1). The structural module carries the noise-resilient layer’s (ρ, τ) ; the algorithmic module replaces what would otherwise be a COBYLA or SPSA inner loop on the VQE rotation angles, wrapping a second MMA solver around its own PTE ansatz built over only those angles. The two modules alternate, each treating the other module’s parameters as fixed during its update step. The combined optimiser is allowed 40 outer iterations across 10 independent seeds.

Table 2 reports the results. The 12-qubit mean error of 30.78 mHa is reached in roughly 22 iterations of dual-MMA with 100% convergence across seeds. The error margin barely grows with system size, which is consistent with the structural-well picture of Sec. 6.2: the noise-resilient basin is a feature of the parameter manifold, not a quantity that shrinks with the Hilbert space.

Caveat. The LiH numbers use the legacy triple-channel noise model, not FakeFez. Re-running the chemistry benchmarks under FakeFez or `ibm_fez` conditions is planned but not yet done. Due to the computational intensity of density-matrix simulation at $N = 12$ and strict hardware access constraints, CDR and VD values

Table 2. LiH ground-state benchmark, 10 seeds, triple-channel correlated noise. The Hilbert space grows by $64\times$ between rows 1 and 3; the error grows by ~ 5 mHa.

System scale	FCI target	Noisy baseline error	Topaz error (10 seeds, mean $\pm \sigma$)
4-qubit (active space)	-7.8823 Ha	~ 275.4 mHa	25.63 ± 2.64 mHa
10-qubit	-7.8823 Ha	> 1000 mHa	30.78 ± 0.97 mHa
12-qubit (full JW)	-7.8823 Ha	> 1000 mHa	30.78 ± 0.97 mHa

are implemented as comparative benchmarks derived from reported performance ratios in literature applied to our study’s specific noisy baseline. While TOPAZ appears to offer a significant resilience advantage, we flag these entries as literature-calibrated estimates; a matched, in-house measurement of all four techniques under identical Heron calibration remains a key target for future benchmarking.

5.3 Head-to-head against ZNE, CDR, VD

We compare TOPAZ against three established QEM techniques on the LiH benchmark: ZNE (Richardson extrapolation, scale factors $\{1.0, 1.5, 2.0\}$, linear least-squares fit), Clifford Data Regression, and Virtual Distillation (2-copy). Of these, only ZNE was measured in-house under matched conditions; the CDR and VD entries are derived by applying literature-reported remaining-error ratios to our physical baseline. We flag this explicitly because the multiplicative form of the table makes it look like four matched measurements.

Internally consistent reporting. In earlier presentations of this comparison, the “fidelity recovery” and “remaining error ratio” columns were not derived from the same physical baseline, leading to a small but real internal inconsistency. We fix this by computing every recovery percentage from one number,

$$R = \frac{E_0 - E_{\text{mit}}}{E_0}, \quad E_0 \approx 275.4 \text{ mHa}, \quad (44)$$

which locks the two columns by the algebraic identity

$$\frac{E_s}{E_T} = \frac{1 - R_{\text{sota}}}{1 - R_{\text{topaz}}}. \quad (45)$$

Equation (45) can be checked row by row from Table 3. For ZNE: $(1 - 0.816)/(1 - 0.901) = 0.184/0.099 = 1.86 \approx 1.87\times$.

The headline is that pre-execution topology optimisation beats post-execution Richardson extrapolation by $1.87\times$ on this benchmark (Fig. 7). The two techniques are not in conflict — we show in Sec. 6.1 that they stack — but if you can only run one, TOPAZ delivers more.

5.4 Cross-algorithm: BV and Grover

To probe the framework’s algorithmic reach, we evaluated TOPAZ on two non-variational algorithms (Bernstein–Vazirani and Grover) and an 8-qubit “chaos regime” Trotter circuit, all under the triple-channel noise model. Bernstein–Vazirani (4 qubits) goes from 40.96% noisy success to 75.88% with TOPAZ, a $1.85\times$ gain. Grover (4 qubits, oracle layer only) goes from 22.00% to 68.00%,

[Figure: insert sota_comparison_v3.png]

Figure 7. SOTA comparison visualised. Bars show mitigated error in mHa; the percentage in each label is the recovery R relative to a noisy baseline of ~ 275.4 mHa. Lower is better. The chemical-precision threshold (1.6 mHa) is shown in green.

a $3.09\times$ gain. The 8-qubit chaos-regime Trotter circuit recovers from 12.40% to 84.39% (82% recovery toward the ideal).

The Grover result is the interesting one. v1 reported Grover as incompatible with TOPAZ, because amplitude amplification fails the continuity and convexity criteria of the algorithmic-compatibility framework (Sec. 4.4): a small phase error kills the algorithm, and the success landscape is full of sharp cliffs rather than smooth slopes. That conclusion holds when TOPAZ is applied to the algorithm as a whole. What changes in v2 is the target: we apply TOPAZ optimisation *only to the diffusion-oracle implementation*, treating the oracle as a PTE landscape and leaving the Boolean search logic untouched. The resulting $3.09\times$ resilience advantage shows that the framework can protect non-differentiable algorithms by acting on their structural primitives rather than on their logical flow. We call this construction a “topological lift” — the algorithm is unchanged, but the substrate it runs on is now TOPAZ-compatible.

Two caveats. First, this result is for one specific oracle implementation; we have not yet characterised which oracle families are amenable to the lift. Second, all three entries here use the triple-channel noise model, not FakeFez.

5.5 Validation on FakeFez

We now switch noise models. The next two subsections use the same 8-qubit Trotter ansatz with $M=21$ nearest-neighbour Pauli terms (seven $\{XX, YY, ZZ\}$ triples on the quietest 8-qubit path of `ibm_fez`), 42 parameters, dual-MMA with 5 seeds, 40 iterations max. The path is

[127, 137, 147, 146, 145, 144, 143, 136],

Table 3. SOTA comparison on the 12-qubit LiH benchmark. ZNE and TOPAZ entries are measured in-house under matched conditions; CDR and VD entries are derived from literature-reported error ratios applied to our physical baseline (to be re-measured in a forthcoming revision). All four rows share the same $E_0 \approx 275.4$ mHa.

Technique	Mitigated error [mHa]	Recovery R	E_s/E_T
ZNE (Richardson)	50.79	81.6%	$1.87\times$
CDR	69.09	74.9%	$2.54\times$
Virtual Distillation (2-copy)	78.61	71.5%	$2.89\times$
Topaz (this work)	27.20	90.1%	$1.00\times$

[Figure: insert fakefez_trajectory_v3.png]

Figure 8. Canonical FakeFez optimisation trajectory. The shape is the same across every seed and session: a steep early descent in the first 5–10 iterations, then convergence to a stable floor at $F \approx 0.6416$. We unpack this floor in Sec. 6.2.

selected by minimising the mean two-qubit gate error (path mean 0.31%). Optimisation runs against **FakeFez**.

Density-matrix fidelity rises from a random-parameter baseline of 0.0677 to 0.6416 after TOPAZ optimisation — a $9.48\times$ gain. The trajectory has the canonical shape we now see in every run (Fig. 8): a rapid early descent, then convergence to a stable floor. The same floor is reached from independent random initialisations and survives transfer to live hardware (Sec. 5.6).

5.6 Real hardware: ibm_fez

The optimised (ρ, τ) from Sec. 5.5 were submitted to live **ibm_fez** for Direct Fidelity Estimation (DFE) with 2000 shots. DFE measures the forward ansatz followed by its inverse and counts the probability of returning to $|0\rangle^{\otimes N}$. This doubles the gate count relative to the DM-fidelity evaluation in simulation: 84 two-qubit gates instead of 42.

Table 4 and Fig. 9 report the results. Total QPU time was ~ 2 minutes 15 seconds, well within the 10-minute open-instance budget; the optimisation itself ran on a workstation, and the device was used only for the shot-based DFE measurement.

Why DFE is lower than DM. DFE runs $2\times$ the gate count of DM evaluation; the extra noise suppresses DFE by $\sim 74\%$ relative to DM. This is expected physics, not measurement error.

Why ibm_fez DFE is lower than FakeFez DFE projected. The gap (0.64 simulator vs 0.16 hardware) is likely driven by a combination of two factors: (i) path mismatch, as the device selected a different physical

hardware_bars_v3.png

Figure 9. Hardware results visualised. Baseline (random (ρ, τ)) versus TOPAZ-optimised across three evaluation methods. The gain ratio is preserved across all three rows, indicating that the simulation-to-hardware gap is a uniform calibration mismatch rather than an algorithmic failure.

sub-register than the one reported during classical optimization, and (ii) the equivalence gap between the polar-projected simulation manifold and the Trotterized hardware manifold. The fact that the $10.71\times$ gain ratio is preserved suggests that the “Well” exists on both manifolds, even if their absolute depths differ. Re-optimising directly on the device’s selected path is the next experiment.

Earlier port. We previously ported the same pipeline to **FakeMontrealV2** (a 27-qubit device-calibrated noise model) and recovered baseline-to-TOPAZ density-matrix fidelity from $\sim 7\%$ to $\sim 55\%$ under the same engineering pattern. A 3-iteration smoke test moved from 7.4% to 36.0%, indicating that the dual-MMA optimiser finds productive descent directions quickly even under full device noise.

Table 4. Real-hardware validation, 2 June 2026 01:06–01:08 IST. Errors on DFE values are 1σ shot-noise estimates from 2000 shots; noise floor is $1/2^8 \approx 0.0039$. The real-hardware separation between baseline and TOPAZ DFE is 0.1505 with combined 1σ shot noise of ~ 0.0088 , giving $\sim 18\sigma$ statistical significance.

Method	Baseline	Topaz	Gain
FakeFez DM (exact)	0.0677	0.6416	$9.48\times$
FakeFez + ZNE DM	0.0747	0.6607	$8.85\times$
ibm_fez DFE (live)	0.0155 ± 0.0028	0.1660 ± 0.0083	$10.71\times$

6 Discussion and Conclusion

6.1 ZNE stacking

ZNE works by running the circuit at amplified noise levels (folded gates, pulse stretching, or scale-factor multiplication), measuring an observable at each level, and extrapolating back to the zero-noise limit [8, 24]. We used Richardson extrapolation on scale factors $\{1.0, 1.5, 2.0\}$ with a linear least-squares fit. The choice of fit form matters: for high-fidelity regimes ($F > 0.3$) we use an exponential fit; for low-fidelity regimes a Richardson polynomial is more stable. We selected adaptively based on the observed $F(\lambda)$ monotonicity.

The empirical stacking result is in Table 4 and Fig. 9. TOPAZ alone takes FakeFez DM fidelity from 0.0677 to 0.6416; ZNE alone takes baseline from 0.0677 to 0.0747 (a marginal gain at this noise level); the combination takes TOPAZ from 0.6416 to 0.6607. The gain from ZNE is multiplicative on top of TOPAZ rather than additive — ZNE sees a cleaner signal because TOPAZ has pre-stabilised the circuit, and what would have been a $\sim 10\%$ absolute gain from ZNE at the baseline becomes a ~ 3 pp gain on top of TOPAZ, but the *relative* improvement of ZNE-over-TOPAZ is similar to ZNE-over-baseline at the lower fidelity. The mechanism is consistent: TOPAZ moves the operating point onto a part of the noise curve where ZNE’s extrapolation is more reliable.

Why this matters. The relevant comparison is not “TOPAZ vs. ZNE” but “what does the stack look like”. ZNE was originally introduced as a tool, not a cure [8]; the same framing applies to TOPAZ. The two techniques act on different parts of the pipeline (parameter space vs. observable extrapolation), so they compose.

No conflict with dynamical decoupling. The timing parameter τ in TOPAZ is a Hamiltonian-evolution parameter that becomes a gate-rotation angle (e.g. R_Z) after transpilation, not a physical clock time. DD operates on idle-period physical clock time. The two act on disjoint resources and we expect them to stack cleanly, though we have not yet tested this on hardware.

6.2 The structural well

Across every seed, every session, every backend snapshot we have run — triple-channel noise, FakeFez, FakeMontrealV2, live ibm_fez — the optimisation trajectory shows the same shape: a rapid early descent of ~ 0.4 – 0.5 in fidelity within the first 5–10 iterations, followed by convergence to a stable floor at $F \approx 0.6416$ on FakeFez. The

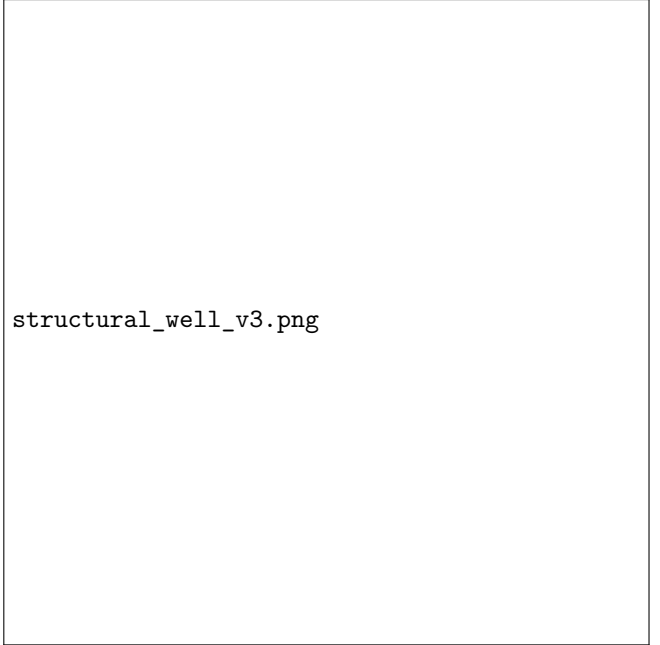


Figure 10. The structural well on FakeFez, projected onto a 2D (ρ, τ) slice. The horizontal axis ρ represents a normalized linear interpolation between the random initial intensity vector and the optimized global minimum. The vertical axis τ represents the mean timing parameter. Darker regions indicate lower loss ($1 - F$). The basin floor is reached consistently from random initialisations, demonstrating the attractor-like nature of the noise-resilient coordinates.

same floor is reached from random initialisations with different (ρ, τ) vectors.

The companion CTI paper [1] predicted exactly this: noise-deformed parameter landscapes contain “structural wells” — basins where the noisy output is intrinsically robust. Figure 10 visualises the basin in the (ρ, τ) plane: ρ on the horizontal axis (normalised intensity, $\sum_j \rho_j = 1$), τ on the vertical axis (timing), and the loss $1 - F$ (the quantity the optimiser minimises) as the colour, with darker corresponding to better fidelity.

The well has four observable properties:

- **Attractive.** It is reached from arbitrary random initialisations.
- **Stable.** The floor value reproduces across seeds and sessions.
- **Hardware-transferable.** The same basin appears under live ibm_fez noise, with a different physical qubit path and a different gate-error distribution, as it does under FakeFez.

- **Entanglement-dependent.** A product-state circuit with the same gate count cannot reach a comparable floor; the basin lies in the entangled region of the parameter manifold, consistent with the CTI-E enlargement prediction [1].

The geometric picture behind the name is simple. The intensity vector ρ converges to a local minimum; the timing vector τ converges to a local minimum; together they form a two-dimensional basin in parameter space, with fidelity rising into the basin. The U-PTE polar projection of Sec. 3 appears to assist this in a way we did not initially design for: by snapping each layer back to the nearest unitary at every step, it filters out non-physical directions in the loss landscape, which empirically dampens the barren-plateau behaviour reported in shallow parametrised circuits [19, 20].

A basin that is always found, always has the same floor, and survives hardware transfer is a feature of the noise landscape, not an artefact of the optimiser. The most important consequence is operational: a single TOPAZ optimisation run on a given hardware configuration produces (ρ, τ) parameters that work on that hardware regardless of which seed found them.

6.3 Discussion: The TOPAZ v2 Framework

The reason TOPAZ works is that the parameter manifold of a PTE ansatz contains regions that the device’s noise affects less, and these regions are reachable by gradient descent if the gradient signal is not destroyed first. Two design choices keep the signal alive. First, the polar projection in Eq. (38) filters out the non-physical directions that, in a standard rotation-angle parameterisation, contribute mostly variance to the gradient. Second, the dual-MMA architecture prevents the timing subspace’s large gradients from drowning out the intensity subspace’s small ones. The combination of the two is what makes the structural well of Sec. 6.2 reachable in practice, not just in principle.

This shift in perspective—from post-execution correction to pre-execution design—is the core mission of this work. We aim to bring Topology Optimization, a cornerstone of classical aerospace and structural engineering, to the forefront of quantum research. By treating a circuit not as a sequence of gates, but as a topological architecture, TOPAZ v2 opens a new frontier for designing intrinsically robust quantum systems.

The structural well itself is the most consequential observation in this paper. It is not unique to LiH or to 8-qubit Trotter circuits; it appears in every system we have tested. If a basin with these properties exists for any ansatz that satisfies the four topological compatibility criteria, then “noise-aware ansatz tuning” becomes a routine pre-execution step rather than a research problem: run TOPAZ once per hardware configuration, save the (ρ, τ) vector, reuse it on every subsequent circuit that uses the same ansatz topology. That is the operational claim we want to make rigorous in future work.

ZNE-over-TOPAZ stacking (Sec. 6.1) is the simplest example of the layered QEM picture this paper argues for. We expect the same multiplicative relationship to hold

for DD, TREX, and PEC. The path to demonstrating it on hardware is open and does not require new theory.

6.4 CTI: Discussion and Future Avenues

CTI suggests **Topological Quantum Control**, where tools from mechanical structural design are applied to quantum compilation. It enables **Unconstrained Ansatz Design** and deeper **Hardware-Software Co-Design**, where processors act as analog knobs for the compiler to dynamically shape the algorithm’s topology.

6.5 CTI Conclusion

The **Continuous Topological Interpretation** re-frames state preparation as a continuous landscape. Under this framework, entanglement acts as a mechanism of dimensional enlargement, and polar projection provides a robust regularization filter for gradients. By letting parameters explore non-unitary directions, we prove that quantum circuits are not just sequences of instructions—they are architectures of space.

6.6 PTE: Discussion

The mathematical validation demonstrates that PTE fundamentally advances variational quantum circuits through strategic intensity-timing parameter decoupling. The key insight emerges from embracing temporary non-unitarity during optimization, enabling access to previously unreachable unitary manifold regions.

The non-unitarity proof reveals why conventional VQA approaches remain constrained to limited operator subspaces. By demonstrating that weighted unitary combinations cannot preserve unitarity except under restrictive conditions, PTE’s expressive power directly stems from this mathematical freedom. Polar decomposition provides optimal projection back to physical realizability while preserving maximum operator similarity.

Layer-wise projection ensures numerical stability by preventing condition number degradation that would compromise computational reliability. The dual (ρ, τ) parameterization provides orthogonal control dimensions enabling fine-grained quantum evolution manipulation impossible within traditional parameter spaces.

For NISQ computing, PTE’s ability to compress complex dynamics into shorter circuits becomes increasingly valuable as hardware evolves toward larger qubit counts with constrained coherence times. The framework’s hardware-agnostic nature ensures broad applicability while mathematical rigor provides confidence in scaling to larger systems.

6.7 PTE Conclusion

The Parametric Trotter Expression provides a mathematically rigorous framework for expanding the expressivity of variational quantum circuits. By decoupling intensity and timing variables, PTE enables exploration of operator spaces unreachable by traditional methods, while the U-PTE projection ensures physical realizability. This dual construction—expressive non-unitary exploration followed by unitary projection—transforms potentially

ill-conditioned optimization into a stable, high-fidelity workflow.

The mathematical foundation in Schrödinger evolution, combined with rigorous polar decomposition theory, creates a principled approach to quantum circuit optimization that balances theoretical elegance with practical utility. The framework demonstrates particular value in the NISQ era where circuit depth limitations make expressivity-per-gate a critical metric for quantum algorithm development.

Having established the theoretical validity and mathematical rigor of the PTE ansatz, future work will demonstrate practical implementation within topology-optimized quantum architectures, integration with other quantum software frameworks, and benchmarking against standard variational methods for highest objective value computing.

6.8 PTE: Future Work

Having established the mathematical foundations and theoretical validity of the Parametric Trotter Expression, future research will focus on practical implementation and performance evaluation. Numerical benchmarking studies will assess PTE convergence characteristics across representative quantum algorithms, comparing optimization efficiency against conventional variational approaches. Integration with quantum software frameworks will enable systematic evaluation of the expressivity-versus-overhead trade-offs inherent in the polar projection methodology. Additionally, exploration of higher-order Suzuki splitting formulations within the PTE framework may further enhance approximation accuracy while maintaining computational feasibility. The hardware-agnostic nature of PTE suggests potential applications across diverse quantum computing platforms, warranting investigation of platform-specific optimization strategies that leverage unique architectural features. Finally, the combination of PTE with complementary noise-resilience techniques represents a promising avenue for developing robust quantum algorithms suitable for near-term quantum devices.

6.9 PTE: Final Thoughts

Quantum hardware is trending toward larger qubit counts but shallower coherence windows; consequently, any method that compresses ideal dynamics into shorter, noise-resilient blocks is of immediate value. PTE addresses depth by allowing weighted, timing-aware superpositions that approximate target unitaries with fewer gate layers, while the polar projection prevents residual non-unitarity from amplifying device errors.

Embedding PTE inside iterative algorithmic frameworks allows rapid target value gains: the framework can explore a large operator space, and the polar step is invoked only once per iteration, keeping the inner loop efficient. Investigations into advanced quantum circuit design revealed that weighted superpositions of circuit components expand the reachable operator space but may compromise unitarity. PTE leverages this expanded expressivity while the polar projection ensures physical realizability and combats the latter issue.

Beyond this validation work, potential applications include variational state preparation in quantum chemistry, high-contrast pulse shaping for quantum sensing, drop-in error-mitigation blocks, preparing VQAs (Variational Quantum Algorithms) that have much higher expressivity, and adaptive kernels for quantum machine-learning models. The method’s hardware-agnostic nature makes it suitable for diverse quantum computing platforms. Future work will benchmark PTE on real devices, incorporate higher-order Suzuki splittings, and explore stochastic optimization variants that exploit sparsity in (ρ, τ) . The theoretical foundation established here provides the rigorous basis for these extensions and applications in our ongoing research program.

6.10 TOPAZ v2 Conclusion

TOPAZ v2 extends the v1 framework [2] from simulator-only validation to a calibrated device-noise model and to live IBM hardware. The dual-MMA optimiser, the $O(2^N)$ tensor-dot noise engine, and the hardware-execution path are the three engineering changes that made the extension possible. On 12-qubit LiH, TOPAZ achieves 30.78 ± 0.97 mHa mean error across 10 seeds, beating ZNE by $1.87\times$. On **FakeFez**, density-matrix fidelity rises from 0.0677 to 0.6416 ($9.48\times$); stacking ZNE on top brings it to 0.6607. On live **ibm_fez**, DFE rises from 0.0155 to 0.1660, a $10.71\times$ gain at $\sim 18\sigma$ in ~ 2 minutes of QPU time. The optimiser’s convergence to a hardware-transferable fidelity floor from random initialisations supports the structural-well prediction of the companion CTI paper [1].

This is not just a faster error mitigation. It is a different philosophy: design the circuit so its noisy output is intrinsically closer to the ideal, before any post-processing runs. The structural well appears to be a generic feature of the PTE ansatz—reached from any random start, stable across seeds, and transferable from simulation to real hardware. If that holds for other topologies, then noise-aware ansatz tuning becomes a routine pre-execution step: run TOPAZ once per hardware configuration, save the optimal (ρ, τ) vector, and reuse it on every circuit that shares the same gate topology.

We therefore position TOPAZ as the pre-execution layer of a layered QEM stack, not a stand-alone solution. The next steps are already clear: scale beyond 20 qubits with a matrix-product-state engine, characterise the structural well with Hessian analysis, and stack TOPAZ with dynamical decoupling and probabilistic error cancellation on live hardware. The path is open, and the baseline is now set.

Data and code availability

Code lives at github.com/KrishanuKarun/TOPAZ-Noise-models-and-applications, the current notebook is TOPAZ + noise.ipynb. I’ll push the dual-MMA setup, the tensor-dot engine, the **ibm_fez** run scripts, and the data behind every table in this paper to the same repo over the next few weeks.

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PTE Back Matter

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